APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 1)

SCENARIO: LOGISTICS & OVERTIME HOURS AUDIT

ASSIGNMENT OVERVIEW:

You are assisting the Operations Director at Apex Retail Group. The warehouse department has experienced increased fulfillment requests. The Operations Director wants to audit overtime hours worked by 10 warehouse staff members during the past month to evaluate fatigue risk and budget overruns.
Standard Max Target Overtime Benchmark: 20.0 hours per worker / month.

DATA RECORD:

Worker Names: `"Worker_A", "Worker_B", "Worker_C", "Worker_D", "Worker_E", "Worker_F", "Worker_G", "Worker_H", "Worker_I", "Worker_J"`

Logged Overtime Hours: `14.5, 22.0, 18.5, 31.0, 12.0, 25.5, 19.0, 28.0, 15.0, 21.0`

PROBLEM 1: DATA CONSTRUCTS & PIPELINE SETUP

In a Jupyter Notebook named `homework_session1_yourname.ipynb`:

1. Store the standard benchmark (20.0) in a variable named `target_benchmark`.
2. Store the 10 overtime hour values in a list named `overtime_hours`.
3. Store the 10 worker names in a list named `worker_names`.

PROBLEM 2: QUANTITATIVE AUDIT & METRICS COMPUTATION

Write Python code to compute:

1. Total overtime hours logged across all 10 warehouse workers `(total_overtime)`.
2. Average overtime hours worked per employee `(avg_overtime)`.
3. Compute individual variances from the benchmark for the first 3 workers `(Worker_A, Worker_B, Worker_C)` by directly accessing list indices.

   Formula: **Variance = Worker Hours - Benchmark**
4. Print clean, formatted outputs showing:
   - Total department overtime hours
   - Average overtime hours
   - Calculated variances for `Worker_A, Worker_B, and Worker_C`

PROBLEM 3: BUSINESS ANALYSIS & STAKEHOLDER MEMO

Write a 150-word business memo to the Operations Director addressing:

1. Is the average department overtime above or below the 20.0-hour safety benchmark?
2. Based on a quick visual review of the data list, does overtime appear evenly distributed across workers, or are specific workers driving up the average?
3. What operational risks (e.g., burnout, budget overrun) should the director address?

SUBMISSION DEADLINE: Prior to Session 2